We shall close this section by stating the important theorem of Weierstrass concerning the approximation of continuous functions by polynomial functions. As would be expected, in order to obtain an approximation within an arbitrarily preassigned $\varepsilon > 0$, we must be prepared to use polynomials of arbitrarily high degree.

5.4.14 Weierstrass Approximation Theorem *Let $I = [a, b]$ and let $f : I \to \mathbb{R}$ be a continuous function. If $\varepsilon > 0$ is given, then there exists a polynomial function p_ε such that $|f(x) - p_\varepsilon(x)| < \varepsilon$ for all $x \in I$.*

There are a number of proofs of this result. Unfortunately, all of them are rather intricate, or employ results that are not yet at our disposal. (A proof can be found in Bartle, ERA, pp. 169–172, which is listed in the References.)

Exercises for Section 5.4

1. Show that the function $f(x) := 1/x$ is uniformly continuous on the set $A := [a, \infty)$, where a is a positive constant.
2. Show that the function $f(x) := 1/x^2$ is uniformly continuous on $A := [1, \infty)$, but that it is not uniformly continuous on $B := (0, \infty)$.
3. Use the Nonuniform Continuity Criterion 5.4.2 to show that the following functions are not uniformly continuous on the given sets.
   (a) $f(x) := x^2$, $A := [0, \infty)$.
   (b) $g(x) := \sin(1/x)$, $B := (0, \infty)$.
4. Show that the function $f(x) := 1/(1 + x^2)$ for $x \in \mathbb{R}$ is uniformly continuous on $\mathbb{R}$.
5. Show that if f and g are uniformly continuous on a subset A of $\mathbb{R}$, then $f + g$ is uniformly continuous on A.
6. Show that if f and g are uniformly continuous on $A \subseteq \mathbb{R}$ and if they are *both* bounded on A, then their product fg is uniformly continuous on A.
7. If $f(x) := x$ and $g(x) := \sin x$, show that both f and g are uniformly continuous on $\mathbb{R}$, but that their product fg is not uniformly continuous on $\mathbb{R}$.
8. Prove that if f and g are each uniformly continuous on $\mathbb{R}$, then the composite function $f \circ g$ is uniformly continuous on $\mathbb{R}$.
9. If f is uniformly continuous on $A \subseteq \mathbb{R}$, and $|f(x)| \geq k > 0$ for all $x \in A$, show that $1/f$ is uniformly continuous on A.
10. Prove that if f is uniformly continuous on a bounded subset A of $\mathbb{R}$, then f is bounded on A.
11. If $g(x) := \sqrt{x}$ for $x \in [0, 1]$, show that there does not exist a constant K such that $|g(x)| \leq K|x|$ for all $x \in [0, 1]$. Conclude that the uniformly continuous g is not a Lipschitz function on $[0, 1]$.
12. Show that if f is continuous on $[0, \infty)$ and uniformly continuous on $[a, \infty)$ for some positive constant a, then f is uniformly continuous on $[0, \infty)$.
13. Let $A \subseteq \mathbb{R}$ and suppose that $f : A \to \mathbb{R}$ has the following property: for each $\varepsilon > 0$ there exists a function $g_\varepsilon : A \to \mathbb{R}$ such that g_ε is uniformly continuous on A and $|f(x) - g_\varepsilon(x)| < \varepsilon$ for all $x \in A$. Prove that f is uniformly continuous on A.
14. A function $f : \mathbb{R} \to \mathbb{R}$ is said to be **periodic** on $\mathbb{R}$ if there exists a number $p > 0$ such that $f(x + p) = f(x)$ for all $x \in \mathbb{R}$. Prove that a continuous periodic function on $\mathbb{R}$ is bounded and uniformly continuous on $\mathbb{R}$.

15. Let f and g be Lipschitz functions on A.
    (a) Show that the sum $f + g$ is also a Lipschitz function on A.
    (b) Show that if f and g are bounded on A, then the product fg is a Lipschitz function on A.
    (c) Give an example of a Lipschitz function f on $[0, \infty)$ such that its square f^2 is *not* a Lipschitz function.

16. A function is called *absolutely continuous* on an interval I if for any $\varepsilon > 0$ there exists a $\delta > 0$ such that for any pair-wise disjoint subintervals $[x_k, y_k]$, $k = 1, 2, \ldots, n$, of I such that $\sum |x_k - y_k| < \delta$ we have $\sum |f(x_k) - f(y_k)| < \varepsilon$. Show that if f satisfies a Lipschitz condition on I, then f is absolutely continuous on I.

Section 5.5 Continuity and Gauges[†]

We will now introduce some concepts that will be used later—especially in Chapters 7 and 10 on integration theory. However, we wish to introduce the notion of a "gauge" now because of its connection with the study of continuous functions. We first define the notion of a tagged partition of an interval.

5.5.1 Definition A **partition** of an interval $I := [a, b]$ is a collection $\mathcal{P} = \{I_1, \ldots, I_n\}$ of non-overlapping closed intervals whose union is $[a, b]$. We ordinarily denote the intervals by $I_i := [x_{i-1}, x_i]$, where

$$a = x_0 < \cdots < x_{i-1} < x_i < \cdots < x_n = b.$$

The points x_i $(i = 0, \ldots, n)$ are called the **partition points** of $\mathcal{P}$. If a point t_i has been chosen from each interval I_i, for $i = 1, \ldots, n$, then the points t_i are called the **tags** and the set of ordered pairs

$$\dot{\mathcal{P}} = \{(I_1, t_1), \ldots, (I_n, t_n)\}$$

is called a **tagged partition** of I. (The dot signifies that the partition is tagged.)

The "fineness" of a partition $\mathcal{P}$ refers to the lengths of the subintervals in $\mathcal{P}$. Instead of requiring that all subintervals have length less than some specific quantity, it is often useful to allow varying degrees of fineness for different subintervals I_i in $\mathcal{P}$. This is accomplished by the use of a "gauge," which we now define.

5.5.2 Definition A **gauge** on I is a strictly positive function defined on I. If δ is a gauge on I, then a (tagged) partition $\dot{\mathcal{P}}$ is said to be δ-**fine** if

$$(1) \qquad t_i \in I_i \subseteq [t_i - \delta(t_i), t_i + \delta(t_i)] \quad \text{for} \quad i = 1, \ldots, n.$$

We note that the notion of δ-fineness requires that the partition be tagged, so we do not need to say "tagged partition" in this case.

A gauge δ on an interval I assigns an interval $[t - \delta(t), t + \delta(t)]$ to each point $t \in I$. The δ-fineness of a partition $\dot{\mathcal{P}}$ requires that each subinterval I_i of $\dot{\mathcal{P}}$ is contained in the interval determined by the gauge δ and the tag t_i for that subinterval. This is indicated by the inclusions in (1); see Figure 5.5.1. Note that the length of the subintervals is also controlled by the gauge and the tags; the next lemma reflects that control.

[†]This section can be omitted on a first reading of this chapter.